

June 26, 2020

① $\mathbb{R}^{\mathbb{N}} \rightarrow$ lin. space of all seq. of real no. $x = (x_k)_{k \geq 0}$.

$S \subset \mathbb{R}^{\mathbb{N}}$ set of sol. of $x_{k+2} = kx_{k+1} + x_k$, $k \geq 0$

$T: S \rightarrow \mathbb{R}^2$, $T(x) = (x_0, x_1)$, $(*) x \in S$.

$S \neq \emptyset$

?

T bij?

S lin. space of finite dim.?

• $x = 0_{\mathbb{N}} \in \mathbb{R}^{\mathbb{N}}$ defined by $x_k = 0$, $(*) k \geq 0$ sol. of D.E. $\Rightarrow S \neq \emptyset$.

• $(*) (x_0, x_1) \in \mathbb{R}^2 \Rightarrow \exists!$ sol. of D.E. with x_0, x_1 as initial values.
 $\Rightarrow T$ bij.

• let $x, y \in S$; $\alpha, \beta \in \mathbb{R}$.

$$z = \alpha x + \beta y$$

$$x \in S \Leftrightarrow x_{k+2} = kx_{k+1} + x_k, (*) k \geq 0 \quad (1)$$

$$y \in S \Leftrightarrow y_{k+2} = ky_{k+1} + y_k, (*) k \geq 0 \quad (2)$$

$$z \in S \Leftrightarrow \underline{z_{k+2}} = \underline{kz_{k+1}} + \underline{z_k}, (*) k \geq 0 \quad (3)$$

$$\underline{z_{k+2}} = \underline{\alpha}$$

$$z = \alpha x + \beta y \Rightarrow \underline{z_{k+2}} = \alpha x_{k+2} + \beta y_{k+2} \quad (4)$$

$$\Leftrightarrow k z_{k+1} + z_k = \alpha (k x_{k+1} + x_k) + \beta (k y_{k+1} + y_k)$$

$$\begin{aligned} \Leftrightarrow k z_{k+1} + z_k &= k (\alpha x_{k+1} + \beta y_{k+1}) + \alpha x_k + \beta y_k \\ &= \alpha (k x_{k+1} + x_k) + \beta (k y_{k+1} + y_k) \\ &= \alpha x_{k+2} + \beta y_{k+2} \quad (5) \end{aligned}$$

$$(4), (5) \Rightarrow z_{k+2} = k z_{k+1} + z_k, (*) k \geq 0$$

$$\Rightarrow z \in S, \text{ indeed } \Rightarrow S \text{ lin. space } (*)$$

T lin. map $\} \Rightarrow T$ isomorphism of lin. spaces $\Rightarrow \dim S = \dim \mathbb{R}^2 = 2$.
 T bij (**)

(*), (**) $\Rightarrow S$ lin. space of finite dim.

same pb, but for: $x_{k+2} + kx_k = \sin k$, $k \geq 0$

Let $x, y \in S$; $\alpha, \beta \in \mathbb{R}$, $z = \alpha x + \beta y$.

$$x \in S \Leftrightarrow x_{k+2} + k \cdot x_k = \sin k, \quad (\forall) k \geq 0$$

$$y \in S \Leftrightarrow y_{k+2} + k y_k = \sin k$$

$$z \in S \Leftrightarrow \underline{z_{k+2} + k z_k = \sin k}$$

$$\Leftrightarrow \underline{\alpha x_{k+2} + \beta y_{k+2} + k(\alpha x_k + \beta y_k) = \sin k}$$

$$\Leftrightarrow \underline{\alpha(x_{k+2} + k x_k) + \beta(y_{k+2} + k y_k) = \sin k}$$

$$\alpha \cdot \sin k + \beta \cdot \sin k = \sin k$$

$\Rightarrow \alpha + \beta = 1$ not always true, $(\forall) \alpha, \beta \in \mathbb{R}$

② Gen. sol. of d.e: $x'' + 2t(x') = 0$.

$$y = x' \Rightarrow \text{eq. becomes: } y' + 2t y = 0 \quad | \cdot e^{t^2} \Rightarrow$$

$$\Rightarrow e^{t^2} \cdot y' + 2t \cdot e^{t^2} \cdot y = 0$$

$$\Rightarrow (e^{t^2} \cdot y)' = 0 \quad | \int$$

$$\Rightarrow y \cdot e^{t^2} = c$$

$$\Rightarrow y = c \cdot e^{-t^2} = x', \quad c \in \mathbb{R}$$

$$\Rightarrow x' = c \cdot e^{-t^2} \quad | \int \Rightarrow x = c \int_0^t e^{-s^2} ds + c_2 \quad \text{S! , not -!}$$

$$\Rightarrow \underline{x = c_1 e^{-t^2} + c_2} \Rightarrow x = c_1 \int_0^t e^{-s^2} ds + c_2; \quad c_1, c_2 \in \mathbb{R}$$

③ Planar system: $\begin{cases} \dot{x} = y(y+3) \\ \dot{y} = -x(y+3) \end{cases}$

a) phase portrait.

$$\begin{cases} y(y+3) = 0 \\ -x(y+3) = 0 \end{cases} \Rightarrow y \in \{0, -3\}$$

$$\text{If } y=0 \Rightarrow -x \cdot 3 = 0 \Rightarrow x = \frac{1}{3} \cdot 0 \Rightarrow (0, 0)$$

$$\text{If } y=-3 \Rightarrow -x \cdot 0 = 0, \quad (\forall) x \in \mathbb{R} \Rightarrow (a, -3), \quad (\forall) a \in \mathbb{R}$$

$$\text{First integral: } \frac{\dot{y}}{\dot{x}} = \frac{dy}{dx} \Rightarrow \frac{-x(y+3)}{y(y+3)} = \frac{dy}{dx} \Rightarrow$$

$$\Rightarrow \underline{y(y+3)dy = -x(y+3)} \Rightarrow -x dx = y dy \quad | \int$$

$$\Rightarrow -\frac{x^2}{2} + c = \frac{y^2}{2}$$

$$\Rightarrow y^2 = -x^2 + c$$

$$\Rightarrow x^2 + y^2 = c, \quad c \in \mathbb{R}$$

$$\mathbb{R}^2 \rightarrow \mathbb{R}, H(x,y) = x^2 + y^2$$

$$H \text{ global f.i. } \Leftrightarrow \frac{\partial H}{\partial x}(x,y)(\dot{x}) + \frac{\partial H}{\partial y}(x,y)(\dot{y}) = 0$$

$$\frac{\partial H}{\partial x} = 2x \quad \frac{\partial H}{\partial y} = 2y \quad (2x)(y(y+3)) + 2y \cdot (-x(y+3)) = 0$$

$$\frac{\partial H}{\partial y} = 2y$$

$$\Leftrightarrow 2x(y^2 + 3y) - 2xy(y+3) = 0$$

$$\Leftrightarrow 2x(y^2 + 3y) - 2x(y^2 + 3y) = 0 \text{ true}$$

$$\Leftrightarrow y^2 + 3y$$

$\Rightarrow H$ global f.i.

Level curves: concentric circles

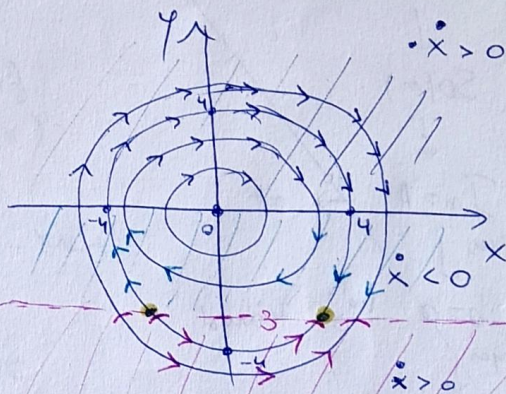
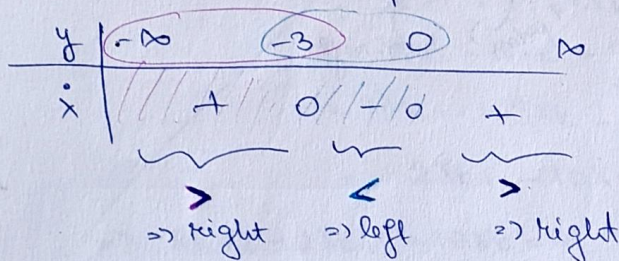
$$x^2 + y^2 = 0 \text{ (origin)}$$

$$x^2 + y^2 = 1 \text{ (unit circle)}$$

Phase portrait: $\rightarrow$ step 1: equil.

$\rightarrow$ step 2: level curves

$\rightarrow$ step 3: arrows



b) Reading the phase portrait, $\lim_{t \rightarrow \infty} \varphi(t, 4, 0)$?
(if $\exists$)
 $\lim_{t \rightarrow \infty} \varphi(t, 1, 0)$?

$\lim_{t \rightarrow \infty} \varphi(t, 4, 0) \rightarrow (4, 0) \in \mathcal{C}(0, 4)$, radius 4, which $\cap$ the line $y = -3$ in 2 points of coord. $(a, -3)$

circle eg: $x^2 + y^2 = r^2$, for origin (center): $(0, 0)$

$$x^2 + (-3)^2 = 4^2$$

$$a^2 + 9 = 16 \Rightarrow a^2 = 7 \Rightarrow a = \pm\sqrt{7}$$

$$\Rightarrow \lim_{t \rightarrow \infty} \varphi(t, 4, 0) = (\sqrt{7}, -3)$$

but $(4, 0)$ is on the RIGHT
 $\Rightarrow$ choose $a = \sqrt{7}$.

$\lim_{t \rightarrow \infty} \varphi(t, 1, 0) \rightarrow (1, 0) \in \mathcal{C}(0, 1)$, radius 1 which $\nmid$ line $y = -3$.

$\Rightarrow \gamma(1, 0)$ closed curve $\Rightarrow \varphi(t, 1, 0)$ periodic fct.

c) $\varphi(t, 4, 0), \varphi(t, 1, 0)$ periodic? $\Rightarrow \nexists$ lim. at ∞ .

$\nexists \lim_{t \rightarrow \infty} \varphi(t, 4, 0) \Rightarrow$ not periodic.

$\nexists \lim_{t \rightarrow \infty} \varphi(t, 1, 0) \Rightarrow$ periodic.

$$O(x) = Ax + b$$

transf? $\det A \neq 0$

yes

no

- proj

is sim?
 $AA^T = I_n$

yes

no

hausd.

$\det A = \pm 1$

$\det A = 1$

$\det A = -1$

$SO(n)$

fixed points
plane $\rightarrow$ refl.
point $\rightarrow$ glide-ref.
point $\rightarrow$ roto-refl.

$I_n = A$ no rot.

yes $b=0$ no transl.

yes id.

(u) $x(t) = \text{temp. coffee}$

$$\dot{x} = k(20-x) = 0 \Rightarrow \underline{c=20=x_p}$$

$$\dot{x} = 20k - xk$$

$k = ?$ s.t. coffee cooled from $100^\circ \xrightarrow[7]{5 \text{ min.}} 80^\circ$

IVP: $\begin{cases} \dot{x} = k(20-x) \\ x(0) = 7 \end{cases}$ linear homogenog. d.e.

$x(t)$ is constant $\Leftrightarrow \dot{x}(t) = 0$

$$x \Rightarrow \underbrace{k(20-x)}_{\neq 0} = 0 \Rightarrow c = 20 \Rightarrow x_p(t) = 20.$$

lin. homog. eq. associated: $\dot{x} + kx = 0 \Rightarrow$ char. eq. method...

$$x = c \cdot e^{-kt} + 20 \quad x_h = c \cdot e^{-kt} \quad \text{gen sol.}$$

$$x(0) = c + 20 = 7 \Rightarrow c = 7 - 20.$$

$$\Rightarrow p(t, 7) = (7-20)e^{-kt} + 20 \xrightarrow[t=100]{100} = 80$$

$$p(5, 100) = 80 \rightarrow (100 - 20)e^{-5k} + 20 = 80$$

$\downarrow$ $\downarrow$ $\downarrow$
 t $t=5$ t
 $\eta = \text{final keep.}$ $\eta = \text{init. keep.}$ η

$$\Rightarrow 80 \cdot e^{-5k} = 60$$

$$\Rightarrow e^{-5k} = \frac{3}{4}$$

$$\Rightarrow -5k = \ln \frac{3}{4}$$

$$\Rightarrow k = -\frac{1}{5} \ln \frac{3}{4}$$

5a). IVP: $\begin{cases} x_{k+1} = -x_k + 3y_k & (1) \\ y_{k+1} = -3x_k - y_k \\ x_0 = \frac{1}{3}, y_0 = 0 \end{cases}$

$\rightarrow$ sol? reduce to second-order d.e.

(1) $\Rightarrow 3y_k = x_k + x_{k+1}$

(2) $\Rightarrow x_{k+2} = -x_{k+1} + 3y_{k+1}$

$$= -x_{k+1} + 3(-3x_k - y_k)$$

$$= -x_{k+1} - 9x_k - 3y_k$$

$$= -x_{k+1} - 9x_k - x_k - x_{k+1}$$

$$= -2x_{k+1} - 10x_k$$

$\Rightarrow x_{k+2} + 2x_{k+1} + 10x_k = 0$ LHDE

$$r^2 + 2r + 10 = 0$$

$$r_{1,2} = -1 \pm 3i$$

$\rightarrow \text{Re}(-1+3i)^k, \text{Im}(-1+3i)^k$

$z = -1+3i \Rightarrow |z| = \sqrt{1+9} = \sqrt{10}$

$\theta \in (0, \pi)$ s.t. $\cos \theta = \frac{\alpha}{|z|} = -\frac{1}{\sqrt{10}}$

$\sin \theta = \frac{\beta}{|z|} = \frac{3}{\sqrt{10}}$

$\Rightarrow z = \sqrt{10}(\cos \theta + i \sin \theta)^k$

$\Rightarrow z^k = (\sqrt{10})^k \cdot (\cos k\theta + i \sin k\theta)$

$\rightarrow \text{Re}(z^k) = (\sqrt{10})^k \cdot \cos k\theta, \text{Im}(z^k) = (\sqrt{10})^k \cdot \sin k\theta$

Let $y = \sqrt{10} \Rightarrow \begin{cases} x_k = c_1 \cdot y^k \cdot \cos(k\theta) + c_2 \cdot y^k \cdot \sin(k\theta) \\ y_k = \frac{1}{3}x_{k+1} + \frac{1}{3}x_k \end{cases}$

$$= \frac{1}{3} \left(c_1 \cdot y^{k+1} \cdot \cos(k\theta + \theta) + c_2 \cdot y^{k+1} \cdot \sin(k\theta + \theta) \right) + \frac{1}{3} \left(c_1 \cdot y^k \cdot \cos(k\theta) + c_2 \cdot y^k \cdot \sin(k\theta) \right)$$

$$y_0 = \frac{1}{3} a \cdot r \cos \theta + \dots \quad \Rightarrow \quad \begin{cases} -\frac{1}{3} + 3a + \frac{1}{3} = 0 \\ a = \frac{1}{3} \end{cases}$$

$$\Rightarrow x_k = \frac{1}{3} r^k \cdot \cos k\theta$$

$$y_k = \frac{1}{9} r^{k+1} \cdot \cos(k\theta + \theta) + \frac{1}{9} r^k \cdot \cos k\theta$$

7b) $\lim_{n \rightarrow \infty} A^{-n} \cdot \begin{pmatrix} 1/3 \\ 0 \end{pmatrix}$, A from a).

$$A = \begin{pmatrix} -1 & 3 \\ -3 & -1 \end{pmatrix}$$

$$X_k = \begin{pmatrix} x_k \\ y_k \end{pmatrix}$$

IVP from a) $\Rightarrow X_{k+1} = A \cdot X_k$, $X_0 = \begin{pmatrix} 1/3 \\ 0 \end{pmatrix}$

sol. of the IVP: $X_k = A^k \cdot X_0$

from a) $\Rightarrow X_k = \begin{pmatrix} \frac{1}{3} r^k \cdot \cos(k\theta) \\ \frac{1}{9} r^{k+1} \cdot \cos(k\theta + \theta) + \frac{1}{9} r^k \cdot \cos k\theta \end{pmatrix}$

Take $k = -n \Rightarrow X_{-n} = A^{-n} \cdot X_0 = \begin{pmatrix} \frac{1}{3} r^{-n} \cdot \cos(n\theta) \\ \frac{1}{9} r^{-n+1} \cdot \cos(n\theta + \theta) + \frac{1}{9} r^{-n} \cdot \cos(n\theta) \end{pmatrix}$

limit.
 $\downarrow$
 $\lim_{n \rightarrow \infty} A^{-n} \cdot \begin{pmatrix} 1/3 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$\lim_{n \rightarrow \infty} r^{-n} \cdot \cos n\theta = 0$ $r = \sqrt{10} > 1$

$\cos n\theta \leq 1$